

Lakshya Agarwal

Full Stack Developer + AI Focus / Virginia Tech (CS + Math , Fall 2028)

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Education

Virginia Tech, *Bachelor of Science in Computer Science*
Minor: Mathematics

Aug 2024 – May 2028 | Blacksburg, VA

Technical Skills

Languages: Python, Java, JavaScript, C/C++, TypeScript, Go • **Frontend:** HTML, CSS, React, Vue.js, T3 Stack, TailwindCSS, Node.js, SwiftUI • **Backend & Cloud:** Google Cloud Platform, MySQL, PostgreSQL, REST API, FastAPI, Django, MongoDB, AWS • **AI:** OpenAI API, Neat-Python, GeminiAPI, Apple Foundation Model

Experience

SDE Assistant, *Virginia Tech IDPRO*

Jan 2026 – Present | Blacksburg, USA

- Developing an **AI-powered** tour guide for **Meta Ray-Ban smart glasses** that identifies paintings, artifacts, and landmarks in **real time**, delivering concise descriptions and interesting facts through bone-conduction audio — **all hands-free**.
- Fine-tuning a Gemini vision-language model** on curated museum and cultural heritage datasets to generate short, contextually rich descriptions from live camera frames, achieving **~2–4 second** response latency per interaction.
- Building a **conversational follow-up system** using **Meta's Android SDK**, allowing users to ask natural language questions about what they're looking at and receive spoken responses **without touching their device**.

Web Developer Intern, *SBM Group Siliguri*

May 2025 – Aug 2025 | West Bengal, India

- Led the frontend redesign for a **real estate company**, migrating from static HTML/CSS to **React.js** improving site performance, responsiveness, and overall user engagement
- Streamlined navigation and property discovery through interactive components and improved site architecture, **contributing to increased website traffic and reduced user drop-off**.
- Conducted competitor benchmarking across real estate platforms to inform UI/UX decisions, **helping drive higher user interaction and lead-generation efficiency**.

Projects

EduBank.AI 📁, • *GO* • *Next.js* • *TailwindCSS* • *UniPDF* • *GeminiAPI*

Feb 2025 – Present

- Building an AI system that generates assessments **exclusively from instructor-provided class notes**, constraining model inputs to reduce hallucinations and off-syllabus content.
- Processes **50–200 pages of course material per class**, enabling instructors to generate **10–50 personalized questions per topic** with adjustable difficulty levels.
- Supports multi-topic courses and concurrent classes while maintaining **100% note-scoped personalization** for each instructor and subject.

Forge 📁, • *Swift* • *GO* • *Apple Foundation Model*

Jan 2026 – Present

- Building an **AI-driven project planning platform** that analyzes team skill profiles and project goals to recommend optimal tech stacks, role assignments, and development workflows.
- Automatically generates **20–60 structured tasks per project**, reducing manual planning time while improving task ownership clarity.
- Dynamically adapts project plans as development progresses, suggesting **feature expansions, task rebalancing, and tooling changes**, improving delivery efficiency for teams

Hackathons

ModelVision - **Top 5 Runner Up** 📁, *Startup Sprint*

Sep 30, 2024

- Enabled **students** to transform **static textbook diagrams** into **interactive 3D models** by pointing their phone at the page, making abstract concepts like molecular structures tangible.
- Reduced student dependency on instructors by integrating an **AI assistant** that answers real-time questions about the 3D model currently in view.
- Cut new subject onboarding time by designing a **modular Unity/Blender pipeline** where **teachers** add content by dropping in assets rather than rebuilding scenes
- Tools Used:** *Unity, Vuforia SDK, Blender, Google Cloud AI*